

# Enhancing Business Process Management (BPM) Sustainability in the Agriculture Sector through Precision Agriculture, IoT, and UAV Technologies

Muhammad Asyraf Bin Rodzoan  
Kulliyah of Information & Communication  
Technology  
International Islamic University, IIUM  
Gombak, Malaysia  
Muhammad.asyraf.rodzoan.ferdin@gmail.com

Asadullah Shah  
Kulliyah of Information & Communication  
Technology  
International Islamic University, IIUM  
Gombak, Malaysia  
asadullah@iium.edu.my

**Abstract**— Agriculture is at the forefront of arguments over global sustainability given that it has a significant impact on resources such land, water, and energy. The significant contributions of business process management, or BPM, to agricultural sustainability is investigated in this article. The agriculture sector may increase operational efficiency, decrease resource waste, and improve sustainability results by integrating BPM with cutting-edge technologies like IoT and UAVs, along with process optimization methodologies. This study addresses the consequences of BPM-driven sustainability, with an emphasis on crop production, resource management, and environmental protection. It does this by using case examples and a review of recent literature. Furthermore, we suggest a paradigm for implementing BPM in agriculture that prioritizes stakeholder engagement and ongoing improvement.

**Keywords**— *Business Process Management (BPM), Sustainability, Agriculture, IoT, UAV, Operational Efficiency, Process Optimization*

## I. INTRODUCTION

Globally, there is growing demand from the agriculture industry to improve output while lessening its negative environmental effects. Population growth, depletion of resources, and climate change all call for a shift to more sustainable agricultural practices [9]. Traditionally used in industry and services, business process management (BPM) provides a methodical way to boost productivity, guarantee responsibility, and cut waste, all of which are in line with agriculture's sustainability objectives [1].

This study looks into how process optimisation and technology-driven solutions can be used to achieve sustainable farming practices through the use of BPM. It addresses the gaps in the literature regarding the use of BPM in agriculture, evaluates how it promotes sustainable operations, and offers a framework for implementing BPM in the sector with an emphasis on long-term sustainability cultivation [10].

## II. LITERATURE REVIEW

### A. BPM in Agriculture

In agriculture, BPM is a relatively new application of an established methodology for process optimisation that

focuses on efficiency, flexibility, and continuous improvement [4]. Few studies have examined how BPM can improve agricultural productivity while addressing environmental sustainability [6]. The definition, measurement, analysis, improvement, and regulation of processes are all addressed in BPM frameworks. The procedures that are involved in the agricultural industry comprise every aspect from planting and irrigation to harvesting and distribution. By using BPM to these tasks, agricultural firms and farmers may simplify business processes, utilise less resources, and increase traceability [3].

### B. Sustainability in Agriculture

In the context of agriculture, sustainability refers to methods that satisfy current demands without endangering the capacity of future generations to satisfy their own. According to the research [12], it entails cutting greenhouse gas emissions, fostering biodiversity, and protecting energy, water, and soil. Waste reduction, resource management, and precision agriculture must change in order to include sustainable practices into agriculture. Adoption of technology is essential for facilitating real-time monitoring, decision-making, and resource optimisation, especially with regard to IoT and UAVs [6].

## III. BPM AND SUSTAINABILITY INTEGRATION IN AGRICULTURE

### A. Process Optimization for Resource Management

BPM provides an organised method for efficiently managing agricultural resources. BPM allows for real-time monitoring and modifications by outlining crucial operations including trash disposal, crop management, and water utilisation [5]. By tracking soil moisture levels, weather patterns, and pest activity, IoT sensors combined with BPM frameworks can reduce resource waste by automatically initiating responses like irrigation or pest treatment [11].

### B. Technology Integration: IoT and UAVs

Use IoT and UAV technologies, that offer real-time information on the health of crops, conditions of the soil, and water requirements, have revolutionised agriculture. To enhance decision-making, these types of technologies can be incorporated into BPM frameworks. UAVs, for example, can

take overhead photos of fields, which BPM systems can then analyse to suggest the best times to sow or harvest [6].

Technology and BPM work together to give a comprehensive picture of the whole agricultural process, through field preparation to handling following the harvest. This raises productivity and aligns agricultural practices with environmental sustainability goals by utilising fewer resources, including water, fertiliser, and pesticides [5].

### C. Continuous Enhancement and Feedback Loops

The focal point of BPM lies in its emphasis on ongoing development through repetitive processes and feedback loops. This can be observed in agriculture as the constant enhancement of irrigation systems, planting techniques, and strategies for pest management [1]. BPM can support farmers in making evidence-based decisions that enhance efficiency and sustainability by consistently monitoring key performance indicators (KPIs) associated with sustainability, such as energy consumption, carbon footprint, and water usage [12].

## IV. FRAMEWORK FOR SUSTAINABLE BPM IMPLEMENTATION IN AGRICULTURE

A structured framework is required for the successful integration of BPM into agricultural practices [5]. The following steps are suggested for the sustainable implementation of BPM within agricultural practices:

- (i) **Process Mapping**  
Determine and document the basic agricultural processes, such as planting, watering, managing pests, and harvesting.
- (ii) **Technology Integration**  
Use IoT devices and UAV technology for real-time data collection and monitoring [6].
- (iii) **Performance Measurement**  
Create KPIs that measure sustainability, such as resource usage (water, energy), crop yield, and waste levels [9].
- (iv) **Data-Driven Decision Making**  
Utilise BPM systems to analyse collected data and provide actionable insights to optimise processes [1].
- (v) **Continuous Monitoring and Improvement**  
Review process performance against sustainability KPIs on a regular basis and make adjustments as needed [5].
- (vi) **Stakeholder Collaboration**  
Engage all key stakeholders, including farmers, agronomists, and policymakers, to facilitate the successful implementation and scaling of BPM projects [6].

## V. CASE STUDY: BPM IN PRECISION AGRICULTURE

A case study demonstrating the use of BPM in precision agriculture demonstrates the usefulness of this strategy. In this case, a farm used BPM to optimise water use and monitor irrigation systems using IoT sensors, which led to a 20%

decrease in water consumption over the course of three growing seasons [5]. Furthermore, UAVs were used to track crop health, which made it possible to apply fertilisers precisely and cut down on the amount of chemicals used by 15% [6].

Through the integration of BPM with cutting-edge technologies, the farm managed to decrease its environmental effect while increasing production. This example shows how BPM can help promote sustainable farming methods [1].

### A. Overview of Agriculture in Malaysia

Malaysia's agriculture sector is an important industry that significantly contributes to employment and GDP. Nevertheless, it encounters various obstacles such as climate change, soil depletion, and lack of workforce. The primary products in this diverse sector include tropical fruits, rice, palm oil, and rubber.

### B. Opportunities and Challenges for Sustainability in Malaysia's Agricultural Landscape

The development of Malaysia's agriculture industry is at a crucial stage, with an increasing interest in adopting sustainable practices. Environmental worries stem from the overuse of chemical pesticides and fertilizers, as well as deforestation for palm oil plantations. Despite these concerns, there are chances to improve sustainability by combining advanced technologies with sustainable farming practices.

### C. BPM in Malaysian Agriculture

The challenges facing Malaysia's agriculture sector can be addressed by using business process management (BPM) to optimise procedures and increase resource efficiency. Precision farming techniques, for instance, can help farmers use water and fertilisers more wisely, minimising their environmental impact while maintaining productivity [8].

### D. Case Examples

By incorporating BPM into sustainable practices, the Malaysian Palm Oil Board (MPOB) is working hard to lessen the environmental effect of palm oil production. This includes programs to enhance waste management and the reduction of greenhouse gas emissions. BPM can maximise every facet of rice production, from planting to harvesting, leading to more efficient use of resources and increased yields. The Malaysian Agricultural Research and Development Institute (MARDI) is investigating the application of BPM in rice cultivation in an effort to promote sustainability.

### E. Policy and Regulatory Framework

The National Agrofood Policy and the Malaysian Sustainable Palm Oil (MSPO) certification are two of the measures the Malaysian government has put in place to promote sustainable agriculture. These policies offer a structure for incorporating BPM with sustainability objectives in the agricultural industry.

## VI. CHALLENGES AND MOVING FORWARD

BPM has the potential to increase agricultural sustainability, but its widespread adoption is hampered by a number of issues. These include as follows:

- (i) **Substantial Initial Investment**

Implementing IoT and UAV technologies entails significant financial requirements, which may not be viable for small-scale farmers [10].

- (ii) **Lack of Technical Expertise**  
Farmers and agribusiness operators may not possess the necessary technical skills to effectively utilize BPM and its associated technologies [9].
- (iii) **Complex Data Management**  
The large quantities of data generated by IoT devices and UAVs necessitate advanced data management and analysis tools, which could pose logistical difficulties [6].

Following this, studies ought to concentrate on generating affordable BPM solutions customised for small-scale agricultural operations, along with establishing instructional schemes that furnish farmers with the essential competencies to exploit BPM innovations [7].

#### CONCLUSION

The Business process management provides a workable route towards sustainable agriculture practices when combined with contemporary agricultural technologies like IoT and UAVs. BPM is a crucial instrument for attaining long-term sustainability in the agriculture industry since it not only promotes resource optimisation but also increases operational efficiency [10]. BPM can assist the agriculture industry satisfy the rising demand for food while reducing its environmental impact by emphasising data-driven decision-making, stakeholder engagement, and continuous improvement [9].

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